

Shouth NLP at SemEval-2025 Task 3: Multilingual Fact-Checking Retrieval Using Contrastive Learning

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Abstract

We present a multilingual fact-checking retrieval system for the SemEval-2025 task of matching social media posts with relevant fact checks. Our approach is based on a fine-tuned multilingual E5 model and in the experimentation with several prefixes and language-specific corpus filtering. The system achieves a Success@10 score of 0.867 on the official test set, although with a big variance between languages. Our analysis discover interesting patterns in cross-lingual transfer, with strong results on Malaysian and Thai languages. We make our code public for further research and development.

1 Introduction

The proliferation of misinformation on social media has created an urgent need for automated fact-checking systems that can efficiently detect and verify false claims, given the immense volume of information shared online (Zhou and Zafarani, 2020). Most automated fact-checking system consist of three steps (Guo et al., 2022): first, the claim detection (is this check-worthy?); then, the evidence retrieval (given a claim, retrieve relevant information to verify it); and lastly and claim verification (given a claim and evidence, define if the claim is true or false). The SemEval-2025 Task (Shahi et al., 2025) addresses the second of these steps by requiring participants to develop evidence-retrieval systems for social media posts, out of a large corpus of 272, 447 fact checks among multiple languages.

In this paper, we present our approach to this task, which uses a fine-tuned multilingual embedding model based on the E5 architecture (Wang et al., 2024). Our system employs a contrastive learning framework to optimize the semantic matching between posts and fact checks. We focus in particular on effective multilingual representation learning and cross-lingual retrieval func-

tions. The code for our system is publicly available on GitHub¹.

Our contributions are the following:

- an analysis of various model architectures and their performance among different languages
- an experimentation of how input prefixes affect retrieval accuracy
- an exploration of the importance of language-specific corpus filtering

Our system achieved an average Success@10 score of 0.867 on the official test set, ranking 20th out of 28 published participants.

2 Background

Automated fact-checking has emerged as a critical tool in combating the spread of misinformation online. The SemEval-2025 task focuses especially on multilingual fact-check retrieval, where the goal is to match social media posts with relevant fact-checks from a large multilingual corpus.

The task involves two main challenges: (1) semantic matching between posts and fact checks, which may be expressed differently despite covering the same claim, and (2) effective cross-lingual retrieval, where a post in one language must be matched with fact checks potentially written in different languages.

Recent advances in multilingual embedding models have shown promising results for cross-lingual retrieval tasks (Feng et al., 2022; Litschko et al., 2022). These models learn a shared semantic space between languages, allowing for effective comparison of text regardless of the source language. In particular, contrastive learning approaches have proven effective for training such models, as they explicitly optimize for similarity

¹https://github.com/sanlares/semeval_2025

between related texts while pushing unrelated texts apart in the embedding space.

Our approach builds upon these advances, utilizing the multilingual E5 model architecture (Wang et al., 2024) as a foundation. This model has demonstrated strong performance on multilingual retrieval benchmarks by combining transformer-based encoders with effective contrastive pre-training. We additionally fine-tune this model on the task-specific data to optimize its performance for fact-check retrieval.

3 System Overview

Our system addresses the challenge of retrieving relevant fact checks from a large corpus of 272,447 documents by implementing a fine-tuned contrastive learning approach based on the multilingual E5 model architecture. The system consists of three main components: (1) a text embedding model, (2) a contrastive learning framework, and (3) a retrieval pipeline.

3.1 Model Architecture

The core of our system is built upon the Multilingual E5 model (Wang et al., 2024) which we improve with a custom architecture designed for fact-checking retrieval. The model consists of:

- A transformer encoder (base E5 model)
- A mean pooling layer for sentence representation
- A dense projection layer with GELU activation
- L2 normalization for cosine similarity optimization

The model processes both posts and fact checks using language-specific prefixes (query: and passage: respectively) to optimize the semantic matching between them.

3.2 Training

We implement a contrastive learning approach using the Multiple Negatives Ranking Loss (MNRL), which can be formulated as:

$$L = -\log \frac{\exp(\text{sim}(x_i, x_i^+)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(x_i, x_j)/\tau)} \quad (1)$$

where:

- x_i represents the anchor text embedding (post)
- x_i^+ represents the positive pair embedding (matching fact-check)
- x_j represents all other samples in the batch (negative pairs)
- τ is a temperature parameter that controls the sharpness of the distribution
- $\text{sim}(\cdot, \cdot)$ is the cosine similarity function

The framework employs in-batch negatives for computational efficiency while maintaining effective contrastive learning. The temperature parameter τ is dynamically adjusted during training to optimize the learning process.

3.3 Retrieval Pipeline

The retrieval process follows these steps:

1. **Language Detection:** The system first identifies the language of the input post.
2. **Corpus Filtering:** The fact-check corpus is filtered to prioritize documents in the same language as the input post.
3. **Embedding Generation:** The model generates embeddings for both the input post and the filtered fact checks.
4. **Similarity Ranking:** The system computes Pearson cosine similarity between the post and fact check embeddings.
5. **Top-K Selection:** The 10 most similar fact checks are retrieved and ranked.

This pipeline is optimized for both monolingual and cross-lingual retrieval, with special consideration for language-specific distinctions in the embedding space.

4 Experimental Setup

4.1 Dataset

The dataset used for this study was derived from the official SemEval-2025 competition data. We divided the training set into training (80%) and validation (20%) subsets while maintaining the language distribution for both sets.

For preprocessing, we utilized the `load.py` file recommended by the competition organizers. Our

approach involved creating rich semantic representations for both posts and fact checks. For posts, we concatenated the OCR-extracted text (ocr column) with the manual transcription (text column) from the posts.csv file. Similarly, for fact checks, we combined the claim and title columns from the fact_checks.csv file.

In addition to fine-tuning our primary model, we conducted extensive experiments with various base models available in the SentenceTransformers framework, testing different prefix combinations that yielded varying performance results.

4.2 Implementation Details

The system was implemented using the following configuration:

- **Base Model:** multilingual-e5-large-instruct
- **Framework:** SentenceTransformers (v3.4.1)
- **Deep Learning Backend:** PyTorch (v2.5.1)
- **Python Version:** 3.12.7
- **Computing Environment:** macOS 15.1 (24B83)

4.3 Training Configuration

The model was trained with the following hyperparameters:

Parameter	Value
Maximum Sequence Length	512 tokens
Pooling Mode	Mean
Embedding Dimension	384
Batch Size	4
Evaluation Batch Size	16
Gradient Accumulation Steps	2
Learning Rate	2e-5
Temperature	0.05
Mixed Precision Training	Enabled

Table 1: Model training hyperparameters

4.4 Evaluation Metrics

The primary evaluation metric for our system was Success@10 (S@10), which measures the proportion of posts for which the correct fact check appears in the top 10 retrieved results. We evaluated the model’s performance in both monolingual and cross-lingual settings:

- **Monolingual:** Retrieval within the same language

- **Cross-lingual:** Retrieval across different languages

During training, we also monitored:

- Cosine similarity between positive pairs
- Cosine similarity between negative pairs
- Training loss convergence
- Validation performance at regular intervals

5 Results

We evaluate our system’s performance on both the validation and test sets, analyzing the impact of different model configurations and language-specific performance. Our system achieved an average Success@10 score of **0.867** on the official test set, ranking 20th out of 28 published participants.

5.1 Official Test Results

The results demonstrate considerable variability in model performance amid languages, with Malaysian (msa) showing the strongest performance at 0.978 and English (eng) and Portuguese (por) showing the lowest at 0.796. This disparity suggests that linguistic factors may play a significant role in retrieval performance.

5.2 Model Configuration Analysis

Table 3 shows the performance comparison of different base models and input prefix configurations on the validation set. The results demonstrate that the multilingual-e5-large-instruct model with appropriate prefixes consistently outperforms other configurations.

5.3 Impact of Input Prefixes

Our experiments revealed that the choice of input prefixes significantly impacts model performance. Using the recommended prefixes on multilingual-e5-large-instruct base model (‘passage:’ and ‘query:’) improved the average Success@10 score by 0.013 points (from 0.821 to 0.834) compared to using no prefixes. This improvement was consistent across all languages, with the largest gains observed in:

- English: +0.022 (from 0.766 to 0.788)
- Spanish: +0.012 (from 0.857 to 0.869)

Language	Success@10	Relative Performance
Malaysian (msa)	0.978	+12.8%
Thai (tha)	0.940	+8.4%
Arabic (ara)	0.912	+5.2%
French (fra)	0.894	+3.1%
Spanish (spa)	0.866	-0.1%
German (deu)	0.858	-1.0%
Turkish (tur)	0.828	-4.5%
Polish (pol)	0.806	-7.0%
English (eng)	0.796	-8.2%
Portuguese (por)	0.796	-8.2%
Average	0.867	–

Table 2: Official test set results of the fine-tuned multilingual-e5-large-instruct model by language. Languages are ordered by Success@10 score. Best performance is shown in **bold**. The model shows strong performance on Malaysian and Thai languages.

Base Model	Posts Prefix	Facts Prefix	Success@10
multilingual-e5-large-instruct (Wang et al., 2024)	passage:	query:	0.834
multilingual-e5-large-instruct (Wang et al., 2024)	–	–	0.821
multilingual-e5-small (Wang et al., 2024)	–	–	0.795
e5-large-v2 (Wang et al., 2022)	–	–	0.758
modernbert-embed-base (Nussbaum et al., 2024)	search_query:	search_document:	0.779
paraphrase-multilingual-mpnet-base=v2 (Reimers and Gurevych, 2019)	–	–	0.736

Table 3: Performance comparison of base models on the validation set. Models are ordered by Success@10 score. Best result is shown in **bold**.

- Portuguese: +0.018 (from 0.781 to 0.799)

These results suggest that appropriate input formatting plays a vital role in optimizing the model’s cross-lingual understanding and retrieval functions.

5.4 Corpus Reduction Analysis

Our experiments showed that performance improves significantly when the fact-check corpus is reduced to include only documents in the same language as the query post. For example, when retrieving from a language-specific subset of approximately 16,000 fact checks (versus the full corpus of 272,447), Success@10 scores improved by an average of 10.2%. For more detailed experiments on corpus reduction with the fine-tuned model, see Table 4 in Appendix A.

6 Conclusion

In this paper, we presented a multilingual fact-checking retrieval system based on fine-tuned contrastive learning with the E5 model architecture. Our approach achieved an average Success@10 score of 0.867 on the official SemEval-2025 test set, with specifically strong performance on non-Latin script languages such as Malaysian and Thai.

Our analysis revealed several key outcomes:

- The base model multilingual-e5-large-instruct consistently outperforms other options.
- Appropriate input prefixes can provide substantial performance gains, improving Success@10 by 0.013 points on average.
- There is considerable variability in model performance between languages, suggesting that language-specific tuning could yield extended improvements.
- Corpus reduction by language dramatically improves retrieval accuracy, highlighting the importance of effective pre-filtering strategies.

Subsequent work could focus on addressing the performance discrepancy among languages, perhaps through language-specific fine-tuning or more sophisticated cross-lingual transfer techniques. Moreover, exploring ensemble methods that combine multiple embedding models might improve retrieval accuracy for challenging languages.

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A Detailed Experimental Results

This appendix presents supplementary experimental results and analyses that complement the main outcomes discussed in the paper.

A.1 Corpus Reduction Analysis

Table 4 shows the impact of corpus reduction on the fine-tuned model’s performance. The results demonstrate that using language-specific subsets of the fact-check corpus significantly improves retrieval performance.

Model	Posts Prefix	Facts Prefix	S@10
multilingual-e5	query:	passage:	0.934
multilingual-e5	–	–	0.933
multilingual-e5	post:	fact check:	0.934
multilingual-e5	This is a post...	This is a fact...	0.936
multilingual-e5	<[language]>	–	0.933
multilingual-e5	The follow-ing...	The follow-ing...	0.934

Table 4: Fine-tuned model performance with different configurations when using a reduced corpus. All experiments were conducted on the validation set with a corpus containing only fact checks in the same language as the query posts, resulting in higher overall scores compared to the full corpus evaluation.

Our experiments showed that performance improves significantly when the fact-check corpus is reduced to include only documents in the same language as the query post. For example, when retrieving from a language-specific subset of approximately 16,000 fact checks (versus the full corpus of 272,447), Success@10 scores improved by an average of 10.2%.

A.2 Fine-tuning Impact

The base model ‘intfloat/multilingual-e5-large-instruct’ (Wang et al., 2024) improved from an average Success@10 of 0.834 to 0.867 when fine-tuned with the training dataset, along with the recommended query and passage prefixes. This improvement was consistent for all languages tested in the competition.